MTH 101

Calculus & Analytical Geometry

Complete Typed Notes

Part 1: Lessons 01–12

Properly Formatted with Tables & Math Mode

October 11, 2026

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1 Lesson 01: Coordinates, Graphs and Lines

Definition: Calculus is the study of continuous rates of change of quantities. It is the study of how various quantities change with respect to (w.r.t) other quantities.

Examples:

- How distance changes w.r.t time
- How time changes w.r.t speed
- How water flow changes w.r.t time

1.1 Real Numbers

Type	Definition / Examples
Natural Numbers	$1, 2, 3, 4, 5, \ldots$
Whole Numbers	$0, 1, 2, 3, 4, \ldots$
Integers	$\ldots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots$
Rational Numbers	Ratios of integers p/q (division by zero ruled out)
Irrational Numbers	Cannot be expressed as ratio of integers

Remark: Why not divide by 0?
If $x \neq 0$, the equation $x/0 = y \Rightarrow x = 0 \cdot y \Rightarrow x = 0$ is contradictory.
If $x = 0$, the equation is satisfied by any number y, so the ratio does not have a unique value. These symbols are undefined.

Historical Note: Pythagoras concluded that the size of a physical quantity must consist of a whole number of units plus some fraction m/n. This idea was shattered in the fifth century B.C. by Hippasus of Metapontum who demonstrated the existence of irrational numbers. He proved that $\sqrt{2}$ is irrational.

Examples of irrational numbers: π, $\cos 19°$, $1 + \sqrt{2}$

1.2 Set Theory

Definition: The collection of well-defined objects is called a set.

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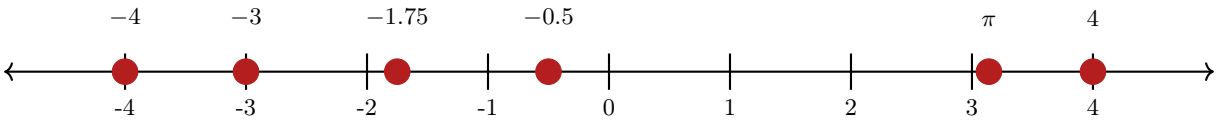
Symbol	Meaning
$a \in A$	a is an element of set A
$a \notin A$	a is NOT an element of set A
$\varnothing$ or $\{\}$	Empty set
$A \cup B$	Union of A and B
$A \cap B$	Intersection of A and B
$A \subseteq B$	A is a subset of B

Set-Builder Notation:

$$\{x : x \text{ is a real number and } 2 < x < 3\}$$

1.3 Coordinate Line

In the 1600's, analytic geometry was developed by the French mathematician Descartes.



One-to-One Correspondence: Each real number corresponds to a single point and each point corresponds to a single real number.

1.4 Order Properties

Definition: For any two real numbers a and b, if $b - a$ is positive, then we say that $b > a$ or that $a < b$.

Property	Rule
(a)	If $a < b$ and $b < c$, then $a < c$
(b)	If $a < b$ and $a + c < b + c$, then $a - c < b - c$
(c)	If $a < b$ and $ac < bc$ when $c > 0$; $ac > bc$ when $c < 0$
(d)	If $a < b$ and $c < d$, then $a + c < b + d$
(e)	If a, b both positive or both negative and $a < b$, then $1/a > 1/b$

1.5 Intervals

Definition: Geometrically, an interval is a line segment on the coordinate line.

Notation	Definition	Type
$[a, b]$	$\{x : a \leq x \leq b\}$	Closed interval
(a, b)	$\{x : a < x < b\}$	Open interval
$[a, b)$	$\{x : a \leq x < b\}$	Half-open
$(a, b]$	$\{x : a < x \leq b\}$	Half-open
$[a, +\infty)$	$\{x : x \geq a\}$	Infinite (closed)
$(-\infty, b]$	$\{x : x \leq b\}$	Infinite (closed)
$(-\infty, +\infty)$	$\mathbb{R}$	Both open and closed

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1.6 Solving Inequalities

Example 4: Solve $3 + 7x \leq 2x - 9$

$$\begin{aligned} 7x &\leq 2x - 12 \\ 5x &\leq -12 \\ x &\leq -\frac{12}{5} \end{aligned}$$

Solution set: $\left(-\infty, -\frac{12}{5}\right]$

Example 5: Solve $7 \leq 2 - 5x < 9$

$$\begin{aligned} 5 &\leq -5x < 7 \\ -1 &\geq x > -\frac{7}{5} \\ -\frac{7}{5} &< x \leq -1 \end{aligned}$$

Solution set: $\left(-\frac{7}{5}, -1\right]$

2 Lesson 02: Absolute Value

Definition: The absolute value or magnitude of a real number a is denoted by $|a|$ and is defined by:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \text{ (non-negative)} \\ -a & \text{if } a < 0 \text{ (negative)} \end{cases}$$

Examples:

$$|5| = 5, \quad \left|-\frac{4}{7}\right| = \frac{4}{7}, \quad |0| = 0$$

2.1 Solving Absolute Value Equations

Example 1: Solve $|x - 3| = 4$

$$\begin{aligned} x - 3 = 4 \quad &\text{or} \quad x - 3 = -4 \\ x = 7 \quad &\text{or} \quad x = -1 \end{aligned}$$

Example 2: Solve $|3x - 2| = |5x + 4|$

Case 1:

$$3x - 2 = 5x + 4$$
$$-2x = 6$$
$$x = -3$$

Case 2:

$$3x - 2 = -(5x + 4)$$
$$3x - 2 = -5x - 4$$
$$8x = -2$$
$$x = -\frac{1}{4}$$

Theorem: For any real number a:

$$\sqrt{a^2} = |a|$$

Definition (Properties): If a and b are real numbers:

Property	Rule
(a)	$\|-a\| = \|a\|$
(b)	$\|ab\| = \|a\|\|b\|$
(c)	$\left\|\frac{a}{b}\right\| = \frac{\|a\|}{\|b\|}$

2.2 Geometric Interpretation

Definition (Distance Formula): On a coordinate line, the distance d between points with coordinates a and b is:

$$d = |b - a|$$

Expression	Geometric Interpretation
$\|x - a\|$	Distance between x and a
$\|x + a\|$	Distance between x and $-a$
$\|x\|$	Distance between x and the origin

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2.3 Inequalities with Absolute Value

Theorem:

Inequality	Equivalent Form
$\|x-a\| < k$	$-k < x-a < k$
$\|x-a\| > k$	$x-a < -k$ or $x-a > k$

Example 1: Solve $|x-3| < 4$

$$-4 < x-3 < 4$$
$$-1 < x < 7$$

Interval: $(-1, 7)$

Example 2: Solve $|x+4| \geq 2$

$$x+4 \leq -2 \quad \text{or} \quad x+4 \geq 2$$
$$x \leq -6 \quad \text{or} \quad x \geq -2$$

Set notation: $(-\infty, -6] \cup [-2, +\infty)$

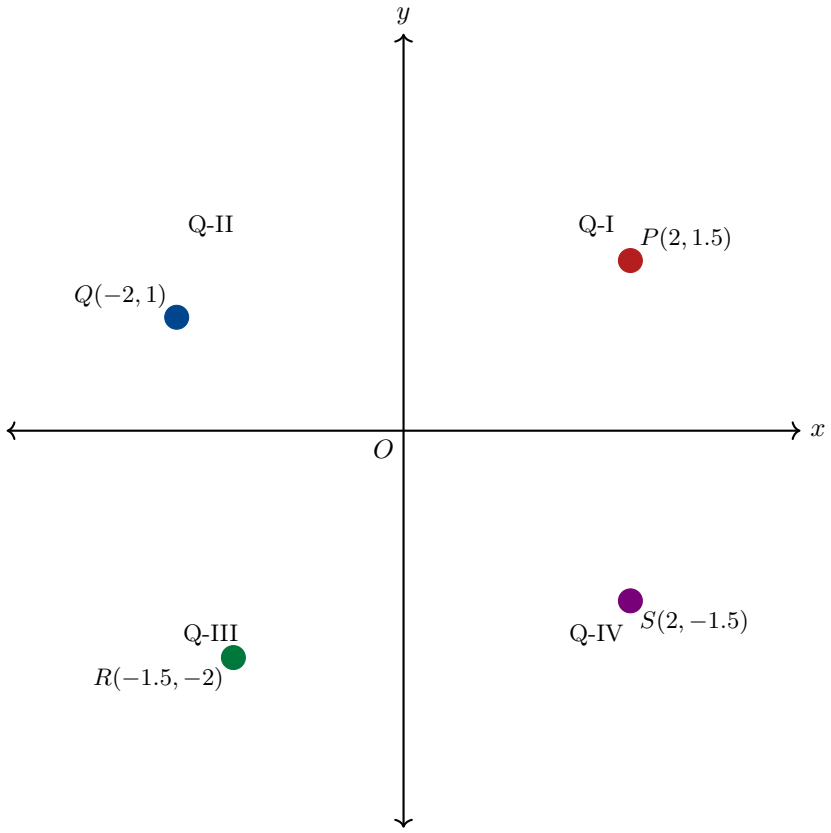
Theorem 1.2.5 (Triangle Inequality): If a and b are any real numbers:

$$|a+b| \leq |a| + |b|$$

3 Lesson 03: Coordinate Planes and Graphs

3.1 The Coordinate Plane

Definition: A plane is just the intersection of two coordinate lines at 90 degrees. It is technically called the coordinate plane.



Definition (Quadrants): The coordinate axes divide the plane into four regions called quadrants.

Quadrant	Signs
Quadrant I	$x > 0,\ y > 0$
Quadrant II	$x < 0,\ y > 0$
Quadrant III	$x < 0,\ y < 0$
Quadrant IV	$x > 0,\ y < 0$

Definition: The graph of an equation in two variables x and y is the set of all points in the xy-plane whose coordinates are members of the solution set of the equation.

Example: Is $(3, 2)$ a solution of $6x - 4y = 10$?

$$6(3) - 4(2) = 18 - 8 = 10 \quad \checkmark$$

3.2 Intercepts

Definition: Points where a graph intersects the coordinate axes are called intercepts.

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- Intersections with the x-axis: $(a, 0)$ — a is called the x-intercept
- Intersections with the y-axis: $(0, b)$ — b is called the y-intercept

Example: Find all intercepts of $3x + 2y = 6$

$$x\text{-intercept (set } y = 0\text{): } 3x = 6 \Rightarrow x = 2 \Rightarrow (2, 0)$$
$$y\text{-intercept (set } x = 0\text{): } 2y = 6 \Rightarrow y = 3 \Rightarrow (0, 3)$$

3.3 Symmetry

Definition: Symmetry describes a graph that is unchanged when certain transformations are applied.

Type	Test
Symmetric about x-axis	(x, y) and $(x, -y)$
Symmetric about y-axis	(x, y) and $(-x, y)$
Symmetric about origin	(x, y) and $(-x, -y)$

Example: $y = \frac{1}{8}x^4 - x^2$
Substituting $-x$ for x:

$$y = \frac{1}{8}(-x)^4 - (-x)^2 = \frac{1}{8}x^4 - x^2$$

Equation unchanged $\Rightarrow$ **Symmetric about y-axis**

4 Lesson 04: Lines

Definition 1.4.1 (Slope): If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are points on a non-vertical line, then the slope m is:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example: Find the slope through:

Points	Computation	Slope
$(6,2)$ and $(9,8)$	$m = \dfrac{8-2}{9-6} = \dfrac{6}{3}$	2
$(2,9)$ and $(4,3)$	$m = \dfrac{3-9}{4-2} = \dfrac{-6}{2}$	-3
$(-2,7)$ and $(5,7)$	$m = \dfrac{7-7}{5-(-2)} = \dfrac{0}{7}$	0

Definition 1.4.2 (Angle of Inclination): For a line L not parallel to the x-axis, the angle of inclination is the smallest angle ϕ measured counterclockwise from the positive x-axis to L.

Theorem 1.4.3: For a non-vertical line:

$$m = \tan\phi$$

If the line is parallel to the y-axis, then $\phi = \pi/2$, so $\tan\phi$ is undefined.

Theorem 1.4.4 (Parallel & Perpendicular):

Condition	Relationship
Parallel	$m_1 = m_2$
Perpendicular	$m_1 m_2 = -1$

Example: Show $A(1,3)$, $B(3,7)$, $C(7,5)$ form a right triangle.

$$m_1 = \frac{7-3}{3-1} = \frac{4}{2} = 2$$
$$m_2 = \frac{5-7}{7-3} = \frac{-2}{4} = -\frac{1}{2}$$
$$m_1 m_2 = (2)\left(-\frac{1}{2}\right) = -1$$

$\Rightarrow AB \perp BC$; thus ABC is a right triangle.

4.1 Equations of Lines

Definition (Point-Slope Form):

$$y - y_1 = m(x - x_1)$$

Definition (Slope-Intercept Form):

$$y = mx + b$$

Definition (General Form):

$$Ax + By + C = 0$$

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where A and B are not both zero.

Example: Line through $(2, 3)$ with slope $-3/2$

$$y - 3 = -\frac{3}{2}(x - 2)$$
$$y = -\frac{3}{2}x + 6$$

Example: Find slope and y-intercept of $8x + 5y = 20$

$$5y = -8x + 20$$
$$y = -\frac{8}{5}x + 4$$

Slope $m = -\frac{8}{5}$, y-intercept $b = 4$

5 Lesson 05: Distance; Circles, Quadratic Equations

Theorem 1.5.1 (Distance Formula): The distance d between two points (x_1, y_1) and (x_2, y_2) in a coordinate plane is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Find distance between $(-2, 3)$ and $(1, 7)$

$$d = \sqrt{(1 - (-2))^2 + (7 - 3)^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Theorem 1.5.2 (Midpoint Formula): The midpoint of the line segment joining (x_1, y_1) and (x_2, y_2) is:

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example: Midpoint of $(3, -4)$ and $(7, 2)$

$$\text{Midpoint} = \left(\frac{3 + 7}{2}, \frac{-4 + 2}{2}\right) = (5, -1)$$

5.1 Circles

Definition (Standard Form of a Circle): The circle of radius r centered at (x_0, y_0) is the set of all points whose distance from (x_0, y_0) is r:

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

Example: Circle of radius 4 centered at $(-5, 3)$

$$(x - (-5))^2 + (y - 3)^2 = 4^2$$
$$(x + 5)^2 + (y - 3)^2 = 16$$
$$x^2 + y^2 + 10x - 6y + 18 = 0$$

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Example: Find center and radius of $x^2 + y^2 - 8x + 2y + 8 = 0$

$$(x^2 - 8x + 16) + (y^2 + 2y + 1) = -8 + 16 + 1$$
$$(x - 4)^2 + (y + 1)^2 = 9$$

Center: $(4, -1)$, Radius: 3

5.2 Quadratic Equations

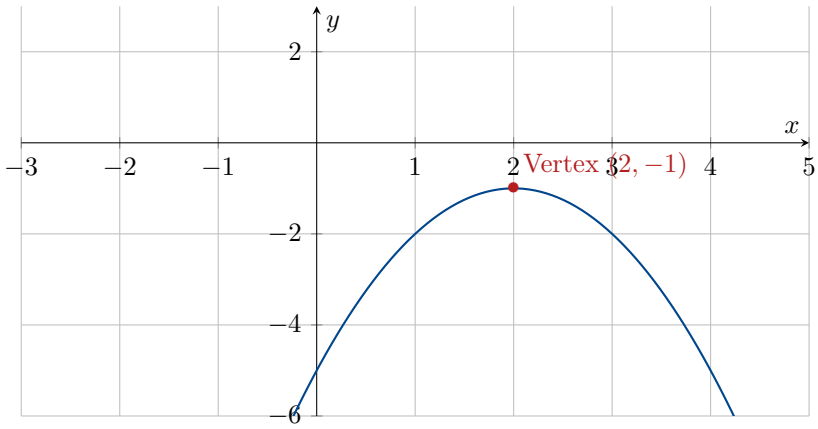
Definition (Quadratic Equation): An equation of the form

$$y = ax^2 + bx + c \quad (a \neq 0)$$

is called a quadratic equation in x. The graph is a parabola.

Definition (Vertex Formula): The x-coordinate of the vertex:

$$x = -\frac{b}{2a}$$



Example: Ball thrown upward: $s = 24.5t - 4.9t^2$

$$\text{Vertex: } t = -\frac{24.5}{2(-4.9)} = 2.5 \text{ sec}$$
$$s = 24.5(2.5) - 4.9(2.5)^2 = 30.625 \text{ m}$$

6 Lesson 06: Functions and Limits

Definition (Function): If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y, we say that y is a function of x.

NOT A Function: $y = \pm\sqrt{x}$ — if $x = 4$, then $y = \pm 2$.

Example: $f(x) = x^2$

$$f(3) = 9, \quad f(-2) = 4$$

Example: $\phi(x) = \dfrac{1}{x^3 - 1}$

$$\phi(1) = \frac{1}{1^3 - 1} = \frac{1}{0} = \text{undefined}$$

6.1 Domain of a Function

Definition (Domain): The set consisting of all allowable values for the independent variable is called the domain of the function.

Example (Natural Domain): $h(x) = \dfrac{1}{(x-1)(x-3)}$

$$\text{Domain: } (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

Example (Restricted Domain): $h(x) = \dfrac{x^2 - 4}{x - 2}$

$$h(x) = x + 2, \quad x \neq 2$$

6.2 Range of a Function

Definition (Range): The set of all possible values for $f(x)$ as x varies over the domain is called the range.

Example: Range of $f(x) = x^2$ is $[0, +\infty)$.

Example: Range of $y = \dfrac{x+1}{x-1}$

$$x = \frac{y+1}{y-1}$$

$y = 1$ is not in the range:

$$\text{Range: } (-\infty, 1) \cup (1, +\infty)$$